RTX 3050: gone from the PCIe bus

What happened on ZRRBITE-PC, what the evidence says, and what to do next, including the BIOS update

ZRRBITE-PC hardware notes

`C:/dev/hardware-notes`

Status as of 2026-11-01 evening: the card is back after the slot test

Where things stand

- From 2026-10-30 16:41 to 2026-11-01 17:31 the RTX 3050 was not detected at power-on. Windows ran on the Intel UHD 770; the monitor on the onboard output.
- Card, HDMI audio function and the CPU's x16 root port were all *phantom* (Device Manager Code 45): absent from the bus, not failing to start.
- 2026-11-01 morning: cold power cycle, card and 8-pin lead reseated. No change.
- 2026-11-01 17:31: card moved to a lower chipset slot, **detected** at Gen3 x4. 17:37: back in the top slot, detected at Gen4 x8. Driver loaded; GPU 1 is back.
- 59 GPU crash reports since 2026-05-04 remain unexplained. The driver update of 2026-10-29 was never the cause.

What it means: the card and the top slot both train again. The fault was in the link path and cleared with the slot change. Its cause is not identified, so it can come back.

Source: diagnosis/gpu-rtx3050-missing-2026-10.md

The machine

What is in the box

Motherboard	Gigabyte Z690 AORUS ELITE DDR4 rev. 1.x, BIOS **F5** (2021-12-17, launch era)
CPU	Intel Core i7-12700K; UHD Graphics 770 integrated, now driving the display
Graphics card	Gigabyte GeForce RTX 3050 8 GB, PCI 10DE:2582, subsystem 1458:4101, rev. A1. Top slot (CPU PCIe x16); the card is natively x8; 130 W, one 8-pin lead
Memory	2 × 16 GB DDR4 (G.Skill F4-4000C18), running at 2666 MT/s, XMP off
Storage	Samsung 970 EVO Plus 500 GB, Samsung 980 PRO 1 TB (Windows)
Network	Realtek RTL8125 2.5GbE on a chipset root port (bus 5, Gen2 x1)
Power supply	**unknown**: label still to be read
OS	Windows 11 Pro for Workstations, build 26200, installed 2026-02-14. Hibernation and Fast Startup off since 2026-09-05, so Shut down is a true power-off
Monitor	Lenovo P34w-20, 3440 × 1440

Source: hardware/inventory.md, scripts/Get-HardwareInventory.ps1

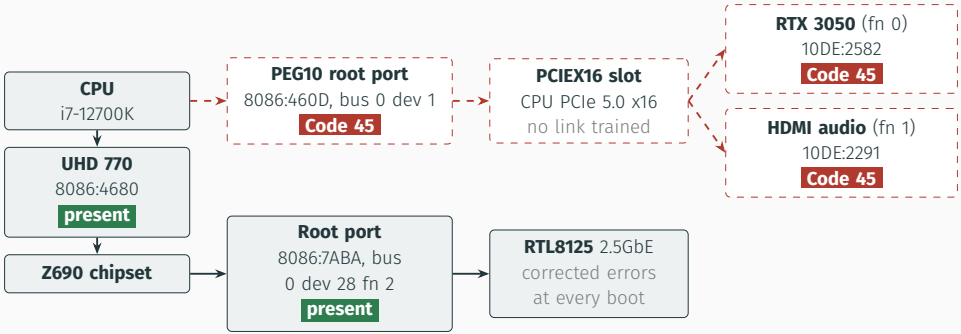
What happened

Timeline

When	Event
2026-05-04	First GPU timeout dump (0x141). Fifteen more in May.
2026-06 to 2026-10	Escalation: black-screen dumps, graphics watchdog dumps, seven blue screens
2026-10-23 04:43	Driver 610.88 arrives via Windows Update; blue screen 0x7E in nvlddmkm the same day
2026-10-29 13:53	Blue screen 0x50 in nvlddmkm
2026-10-29 17:48	Driver 621.01 installed via NVIDIA App. Clean install, no firmware update. Card runs 22 h
2026-10-30 16:12:22	System hangs. Last log write; no crash dump
2026-10-30 16:41:18	Power button held to force off (Kernel-Power 41 carries the timestamp). Boot at 16:41:44: card, audio, root port absent
2026-10-30 16:43:13	Second power-button hold; boot at 16:44:00, same result. Monitor moved to onboard HDMI
2026-10-31	Software diagnosis, SSH set up as a lifeline, vendor report drafted
2026-11-01 08:48 to 09:00	Cold power cycle, card and 8-pin reseated, inspection clean. Two clean boots, card still absent

Source: System log (Kernel-Power, Kernel-General), WER ReportArchive, NVIDIA setup log

The signature: three devices missing at once



The root port belongs to the CPU, not the card; the firmware exposes it only when something trains a link in that slot at POST. Three phantoms, no removal timestamps: *card not detected at POST*.

Source: scripts/Get-GpuState.ps1 on 2026-11-01; last recorded link on 2026-10-29 was Gen4 x8, no AER errors

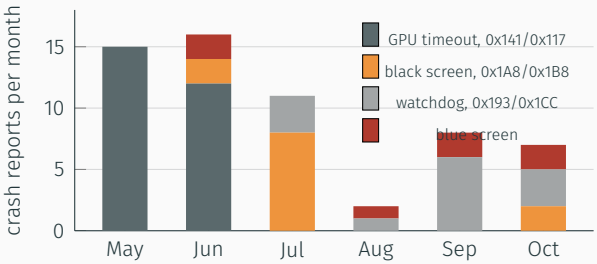
Why it is not the driver

- The root port is enumerated by the BIOS before Windows loads. No driver runs yet.
- The three devices carry no removal timestamp: not disabled or hot-removed in a session, simply absent at the boots of 16:41 and 16:44 and at every boot since.
- 621.01 installed cleanly: no firmware or GOP update component, no reboot required, and the card ran normally for 22 hours afterwards.
- Faults span four driver packages over six months (595.79, 610.74, 610.88, 621.01).
- The 621.01 driver package and its display container service are still installed; `nvidia-smi` reports no device.

What it means: rolling back or reinstalling a driver cannot bring back a device the firmware does not see. Driver work only matters after the card reappears, and then as a DDU clean install.

Source: drivers/nvidia.md; NVIDIA LOG.setup.exe.log; Kernel-PnP and UserPnp events

Six months of warnings



59 kernel reports from 2026-05-04 to 2026-10-31, all in the graphics stack: 52 live kernel events and 7 blue screens.

- **May to June:** recoverable GPU timeouts; the driver reset the GPU and carried on
- **July:** black-screen diagnostics; the display dropped out
- **August to October:** kernel crashes in GPU memory-management code
- **30 October:** the card stops enumerating

The two October black-screen dumps (2026-10-31) are from the integrated GPU, after the card was gone.

Source: C:/ProgramData/Microsoft/Windows/WER/ReportArchive; scripts/Get-CrashHistory.ps1

The seven blue screens

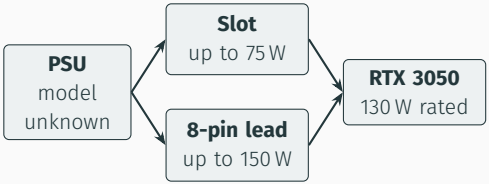
When	Code	Where it crashed
2026-10-29 13:53	0x50	nvlddmkm!_clientUnmapInterBackRefMappings (NVIDIA kernel driver)
2026-10-23 18:35	0x7E	nvlddmkm!refRemoveInterMapping
2026-09-26 09:28	0x7E	dxgmms2!VIDMM_GLOBAL::UnlockAllocation (DirectX video memory manager)
2026-09-24 00:06	0x7E	dxgmms2!VIDMM_GLOBAL::MakeOneVirtualAddressRangeNotResident
2026-08-24 12:20	0xA	nvlddmkm!CNvLAllocReference::unlinkFromAllLists
2026-06-06 23:07	0xA	nt!MiCrashdumpRemovePte, while writing a crash dump
2026-06-06 12:58	0x133	DPC watchdog, nvlddmkm!CNvLAdapter::lockallSubmissions

What it means: five of seven are inside the NVIDIA driver or DirectX's handling of the card's memory. The signatures change from release to release; the component that fails does not.

Source: WER ReportArchive folder names and Report.wer buckets; Application log, Windows Error Reporting 1001

What the hardware is telling us

Power and fans



No software view of the card's power exists while it is off the bus. The 12 V rails can be checked at a spare connector with a multimeter (ATX tolerance ±5 %) or a PSU tester; HWMonitor only sees board sensors.

Observed while the card is undetected

- Both fans spin for a few seconds at power-on, then stop: the firmware's fan self-test, then fan-stop because the chip is cold. A healthy card does the same.
- One fan occasionally restarts at idle, and the heatsink gets hot with no load: without a driver the GPU sits at boot clocks and burns tens of watts.
- A card with no 12 V, or a dead core, would show neither.

What it means: the card is powered and alive at the firmware level. The fault is in PCIe link training, on the card or on the board's x16 port.

Source: owner observations 2026-10-31 and 2026-11-01; hardware/motherboard.md, PSU measurement notes

What has been ruled out

Hypothesis	Test	Verdict
Driver 621.01 broke it	Root port is hidden by firmware before Windows loads; clean install log; 22 h of normal operation	ruled out
PSU dropped out	Kernel-Power 41 events carry power-button timestamps; no power-loss or crash shutdowns in the log	ruled out
Card or cable loose	Reseated card and 8-pin on 2026-11-01; connectors and board inspected; no change	ruled out
8-pin not delivering	Fans self-test at power-on; heatsink warms. A card without auxiliary power would refuse to start	ruled out
Link degraded before failure	Last recorded link Gen4 x8; no AER errors on the card; WHEA errors only on the network card's port	ruled out
Monitor or Windows install	Integrated GPU drives the same monitor at full resolution	ruled out
Card's PCIe interface (PHY or BGA joints)	Card in a chipset slot, then another PC	hypothesis
Board's CPU x16 port, slot or CPU lanes	Card in a chipset slot; BIOS link-speed test; F35 flash	hypothesis

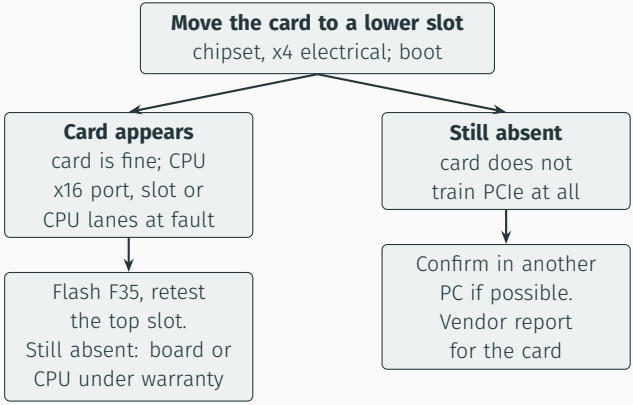
Source: diagnosis/gpu-rtx3050-missing-2026-10.md, Evidence and Ruled out sections

What remains: card or board

First, without opening the case

1. Enter setup (Del). Look for the card anywhere in the device listing.
2. Top slot link speed: Gen3, then Gen1.
3. Load Optimized Defaults (F7), Initial Display Output = IGFX, save (F10).
4. Boot, run `scripts/Get-GpuState.ps1`.

Run 2026-11-01: the card appeared in both slots. Neither branch applied; the link was marginal, not dead.



Source: diagnosis note, Procedure steps 3 and 4; hardware/motherboard.md for the slot layout

The procedure and where we are

	Step	Status
1	Cold power cycle: PSU off, power button held, wait, power on, watch the fans	done 2026-11-01
2	Reseat the card and the 8-pin lead; inspect contacts, slot, board and pins; read the PSU label	done 2026-11-01, PSU label pending
3	BIOS settings: link speed Gen3 then Gen1, Optimized Defaults. Overtaken by step 4	skipped
4	Card or board: card in a chipset slot. Result: detected there, and again in the top slot	done 2026-11-01
5	If the card returns: block automatic driver installs, watch for a week, DDU clean install	active
6	If it does not, or vanishes again: finish the vendor report (PSU, serial, purchase) and submit	on hold

What it means: steps 1 and 2 told us the card was powered. Step 4 brought it back without saying why. Step 5 is where we are.

Source: diagnosis/gpu-rtx3050-missing-2026-10.md, Procedure and Log

Updating the BIOS

Why F5 to F35, and where it fits

What changed between F5 (2021-12) and F35

- **F23:** fix for a blank POST display with GeForce RTX 30-series cards
- **F32:** Intel CSME firmware update. One-way: a BIOS older than F32 cannot be flashed back afterwards
- Microcode, memory compatibility and stability updates throughout

Gigabyte allows the direct jump. Re-read the release notes on the support page before flashing and confirm nothing says “update to Fxx first”.

The image, already downloaded and checked

- `C:/dev/mobo/bios/…/Z690AORUSELITEDDR4.F35`, 32 MiB
- SHA-256 `55d90960…3bc3ced00`
- Contains the board string `Z690 AORUS ELITE DDR4`
- Not committed to the repo

Where it fits

The card ran for years on F5, so the flash alone is unlikely to revive it. Right once the slot test points at the board; harmless either way.

Source: bios/update-bios.md; Gigabyte support page for Z690 AORUS ELITE DDR4 rev. 1.x

Before you flash

1. **BitLocker.** In an elevated terminal (Win+X, Terminal (Admin)): `manage-bde -status C:`
   If protection is on, have the recovery key to hand (Microsoft account, Devices) and suspend it for one reboot: `manage-bde -protectors -disable C: -RebootCount 1`
   A BIOS change alters the TPM measurements; without this Windows may demand the key.
2. **Record the non-default settings.** As of 2026-10-31: boot order (980 PRO) and Initial Display Output. XMP is off.
3. **Power.** Steady mains, no storm. A power loss mid-write leaves the board unbootable; Q-Flash Plus is the recovery path.
4. **Second device** with these notes; the machine is offline for a while.
5. **USB stick**, FAT32, otherwise empty. Copy `Z690AORUSELITEDDR4.F35` to its root.

What it means: do not flash from Windows (GIGABYTE Control Center, @BIOS), and never use a file for the DDR5 board or rev. 2.x. The file name must say DDR4.

Q-Flash, step by step

1. Shut down. Stick into a **rear** USB port, not a front-panel hub.
2. Power on. Press **End** at the Gigabyte logo for Q-Flash, or **Del** for setup and then **F8**.
3. **Update BIOS**, choose the stick, choose `Z690AORUSELITEDDR4.F35`, confirm.
4. **Wait.** The screen may go dark and the machine may restart more than once (CSME update). Touch nothing until setup or Windows appears.
5. Enter setup. **Load Optimized Defaults** (F7), **Save & Exit** (F10).
6. Enter setup again. Re-apply boot order and Initial Display Output. Leave XMP off unless you decide otherwise, and record it if you do.
7. Boot Windows. Enter the BitLocker key once if asked. If you suspended protection, re-enable: `manage-bde -protectors -enable C:`
8. Verify and record: `Get-CimInstance Win32_BIOS | Select SMBIOSBIOSVersion, ReleaseDate` then update `hardware/inventory.md` and the log in `bios/update-bios.md`.

What it means: about fifteen minutes. The only step with a failure mode is the wait in step 4: a restart during the flash is normal, a power cut is not.

Source: bios/update-bios.md

If it will not POST afterwards: Q-Flash Plus

- This board family can flash from USB without a working BIOS.
- Rename the image to `GIGABYTE.bin`, put it alone on a FAT32 stick.
- Plug the stick into the rear port marked for Q-Flash Plus (check the manual for the exact port).
- System off, PSU switched on. Press the Q-Flash Plus button on the rear panel.
- Wait for the LED to stop blinking, several minutes. Then power on normally.

What it means: a failed flash is recoverable on this board: an inconvenience, not a dead motherboard.

Source: bios/update-bios.md; Gigabyte Z690 AORUS ELITE DDR4 manual

Next

Next steps

1. **Leave the card in the top slot.** No automatic driver installs: decline what NVIDIA App and Windows Update offer.
2. **Watch it:** `scripts/Get-GpuState.ps1` after each boot for a week. Code 45 again, or a link below Gen4 x8 under load, means the fault is back.
3. **Crash history:** `scripts/Get-CrashHistory.ps1 -Since 2026-11-01` weekly. If the marginal link caused the May to October timeouts, they stop now.
4. **Clean driver install** per `drivers/nvidia.md` after a clean week.
5. **Flash F35** at leisure. Optional now; worth doing before any new card.
6. **Record** the PSU label and which lower slot was used. Keep the vendor report as a draft; do not submit while the card works.

What it means: every result goes in the Log table of the diagnosis note, dated. The vendor report is built from that table.

Sources and tools

- `diagnosis/gpu-rtx3050-missing-2026-10.md`: symptoms, hypothesis, evidence, ruled-out table, procedure, log
- `hardware/inventory.md`, `hardware/motherboard.md`: what is in the box; slots, firmware behaviour, rail tolerances
- `bios/update-bios.md`: the Q-Flash procedure and its log
- `drivers/nvidia.md`: driver versions, cached installers, clean-install plan
- `scripts/Get-GpuState.ps1`: presence, problem code, last arrival, PCIe link of the card and root port
- `scripts/Get-CrashHistory.ps1`: Kernel-Power 41 decoded, WHEA errors, WER kernel report history
- `scripts/Get-HardwareInventory.ps1`: board, BIOS, CPU, RAM, disks, video
- Vendor report (Claude Doc): `https://claude.ai/artifact/Vq7LVe2ABxcKtUVYbMFStL`
- Gigabyte support: `https://www.gigabyte.com/Motherboard/Z690-AORUS-ELITE-DDR4-rev-1x/support`

Source: all data collected on the affected machine, 2026-10-31 and 2026-11-01, without elevation